

DODGE COUNTY AP, WI, USAWMO: **726509**Lat: **43.426N**Lon: **88.703W**Elev: **936**StdP: **14.21**Time Zone: **-6.00 (NAC)**Period: **97-19**WBAN: **04898****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-6.2	-1.4	-15.2	2.5	-4.0	-10.6	3.2	0.2	24.1	26.7	21.0	26.1	6.2	280	0.544

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.4	87.8	73.3	84.2	71.4	82.0	70.0	77.1	83.5	74.8	80.9	73.0	79.0	11.0	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
75.0	135.9	80.6	72.8	125.9	78.3	71.6	120.8	77.3	41.3	83.0	38.9	82.0	37.2	78.6	88.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.3	19.7	17.6	-13.6	91.8	6.3	2.9	-18.2	94.0	-21.9	95.7	-25.5	97.3	-30.1	99.4
			-14.1	79.5	6.1	3.5	-18.5	82.0	-22.1	84.0	-25.5	86.0	-30.0	88.5
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	46.9	19.7	22.8	34.1	46.0	57.4	66.8	70.9	68.8	62.6	49.9	37.5	24.7
	DBStd	20.18	11.85	11.37	11.29	9.03	8.40	6.65	5.30	5.17	7.82	9.02	9.79	11.26
	HDD50	3710	941	761	510	178	25	0	0	0	5	116	389	785
	HDD65	7161	1406	1181	959	572	265	58	11	22	137	475	826	1249
	CDD50	2574	0	1	17	59	253	503	648	583	382	113	13	2
	CDD65	549	0	0	1	3	29	111	194	140	64	7	0	0
	CDH74	4292	0	0	6	44	272	913	1549	943	508	57	0	0
	CDH80	1047	0	0	0	5	56	242	420	197	120	7	0	0
(14) Wind	WSAvg	7.7	8.3	8.4	8.6	9.5	8.5	7.0	5.9	5.4	6.6	7.8	8.4	8.2
(15) Precipitation	PrecAvg	33.4	1.4	1.3	1.8	3.3	3.8	5.2	4.2	3.5	3.1	2.5	1.9	1.5
	PrecMax	41.7	3.0	3.0	4.8	6.5	9.8	13.0	9.3	11.4	6.1	5.6	6.0	3.5
	PrecMin	26.4	0.2	0.0	0.2	0.6	1.0	1.1	1.4	1.4	0.5	0.6	0.3	0.2
	PrecStd	4.2	0.7	0.7	1.3	1.5	2.0	2.5	2.0	2.4	1.4	1.5	1.2	0.8
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	47.4	55.3	72.7	80.5	85.8	90.3	91.3	88.5	88.2	80.8	67.6	57.0
		MCWB	44.3	50.3	61.3	61.1	67.4	71.7	74.7	75.3	72.3	69.4	56.6	54.9
	2%	DB	41.1	46.3	63.1	72.6	81.1	85.9	87.9	84.4	83.0	73.5	61.1	49.6
		MCWB	37.9	42.5	55.6	58.7	65.7	70.7	74.8	72.7	70.0	63.3	53.3	46.6
	5%	DB	37.2	41.1	55.5	66.1	76.7	82.2	84.1	81.8	80.3	69.9	56.5	43.0
		MCWB	35.3	37.8	49.3	54.5	63.0	69.3	72.3	70.9	68.0	61.9	50.6	40.4
	10%	DB	35.5	36.9	50.1	61.4	72.5	79.6	81.7	79.4	76.0	64.3	52.3	38.5
		MCWB	34.1	34.7	44.5	51.0	61.2	67.4	70.8	68.5	65.8	57.7	47.1	36.3
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	44.6	52.0	63.5	64.6	72.0	78.3	82.3	78.6	75.9	72.0	60.2	55.2
		MCDB	45.8	54.4	66.8	74.3	81.0	84.5	85.7	84.5	82.0	78.5	65.7	56.5
	2%	WB	38.3	43.4	57.1	60.2	68.6	75.2	77.9	75.6	73.1	66.9	55.3	47.3
		MCDB	40.3	45.9	61.8	69.2	76.8	81.4	84.2	81.8	79.9	71.5	59.6	48.9
	5%	WB	35.9	38.4	51.3	56.7	66.1	72.2	75.1	73.7	70.4	63.1	51.6	40.7
		MCDB	37.6	40.7	55.0	65.1	73.5	79.5	81.2	79.1	76.7	66.9	55.7	42.7
	10%	WB	33.4	35.3	44.5	52.6	63.3	69.5	73.2	71.8	68.0	58.8	47.0	36.6
		MCDB	34.9	37.2	49.1	60.3	69.3	76.6	79.0	76.7	73.9	63.9	51.3	38.4
(35) Mean Daily Temperature Range	5% DB	MDBR	13.1	14.3	15.5	18.3	19.1	18.3	18.4	17.8	19.7	17.2	14.2	12.4
		MCDBR	13.5	16.7	22.5	26.8	24.1	21.4	20.2	19.6	22.9	22.9	20.6	15.3
		MCWBR	11.7	13.3	15.7	15.7	12.6	11.2	10.7	10.6	11.5	13.5	14.8	13.2
	5% WB	MCDBR	12.7	15.5	20.6	24.0	20.7	19.3	17.8	17.1	19.6	19.5	19.2	14.3
		MCWBR	11.3	12.7	16.0	15.9	12.3	12.0	11.6	11.1	11.7	14.8	15.5	13.2
(40) Clear-Sky Solar Irradiance	taub		0.275	0.316	0.334	0.377	0.399	0.415	0.410	0.400	0.370	0.333	0.295	0.274
	taud		2.386	2.260	2.362	2.285	2.273	2.267	2.281	2.299	2.376	2.473	2.509	2.433
	Ebn at Noon		274	277	287	281	277	272	272	271	272	270	266	264
	Edh at Noon		24	33	34	40	42	42	41	39	33	27	22	21
(44) All-Sky Solar Radiation	RadAvg		517	852	1198	1443	1706	1905	1937	1665	1354	864	558	401
	RadStd		72	89	137	131	134	137	90	105	112	98	46	38

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	-1.40	-3.71	-4.34	N/A	-3.02	-3.11	N/A	N/A	N/A	N/A	N/A
(47) Regional (56 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page